

Rewriting the Line Filter with an Isotropic WVF Pair

A derivation showing where the LF can be simplified without making the WVF anisotropic.

1 The Constraint

The simplification has to keep the Wide View Filter isotropic. That means the polynomial fit used to estimate first derivatives must use one fixed neighborhood around each pixel, independent of the candidate LF orientation. The orientation-dependent part is allowed to live in the line averaging stage, but the WVF itself should not be rebuilt as a different rotated support for every θ .

This distinction matters because the standard LF is usually written as if a WVF is evaluated at many virtual points for every candidate line angle [1], [2]. If that statement is implemented literally, it looks like the algorithm needs a different WVF derivative row for every θ . The derivation below shows that this is unnecessary when the WVF neighborhood and polynomial basis are isotropic. The normal derivative needed by a line at angle θ is exactly a sine-cosine combination of one canonical G_x row and one canonical G_y row.

2 The Isotropic WVF Pair

Let $P = \{p_i\}_{i=1}^{N_p}$ be a fixed circular set of neighbor offsets around a pixel x . The key word is fixed. The same offsets are used no matter which LF angle is being tested. Let $\mathbf{A}$ be the Taylor design matrix evaluated at those offsets, and let $\mathbf{A}^+$ be its pseudoinverse. For a local vector of sampled image intensities $\mathbf{b}_x$, the isotropic derivative estimates are

$$G_{x(x)} = \mathbf{e}_x^\top \mathbf{A}^+ \mathbf{b}_x, \quad G_{y(x)} = \mathbf{e}_y^\top \mathbf{A}^+ \mathbf{b}_x. \quad (2.1)$$

Here $\mathbf{e}_x^\top \mathbf{A}^+$ and $\mathbf{e}_y^\top \mathbf{A}^+$ are the first-derivative rows of the same local polynomial fit. They are not orientation-indexed. They are just two fixed filters. In implementation terms, G_x and G_y are obtained by two image correlations with two canonical WVF derivative kernels.

Now introduce a candidate line direction

$$\mathbf{v}_\theta = (\cos \theta, \sin \theta)^\top. \quad (2.2)$$

The edge-normal direction associated with this line is perpendicular to $\mathbf{v}_\theta$:

$$\mathbf{n}_\theta = (-\sin \theta, \cos \theta)^\top. \quad (2.3)$$

The normal derivative of the same fitted polynomial is

$$D_{n,\theta} I(x) = \mathbf{n}_\theta \cdot \begin{bmatrix} G_{x(x)} \\ G_{y(x)} \end{bmatrix} = -\sin \theta G_{x(x)} + \cos \theta G_{y(x)}. \quad (2.4)$$

This is the central identity. It follows from the chain rule. If the local normal coordinate u is defined by $u = -X \sin \theta + Y \cos \theta$, then

$$\frac{\partial}{\partial u} u = -\sin \theta \frac{\partial}{\partial X} X + \cos \theta \frac{\partial}{\partial Y} Y. \quad (2.5)$$

Since the Taylor basis contains the complete polynomial terms up to the selected degree, rotating coordinates changes the parameterization of the same fitted polynomial space. It does not require a new anisotropic WVF support. The first derivative in the normal direction to the line is therefore exactly the $-\sin \theta, \cos \theta$ combination in Equation 2.4.

The LF normal derivative equals one isotropic Gx/Gy pair

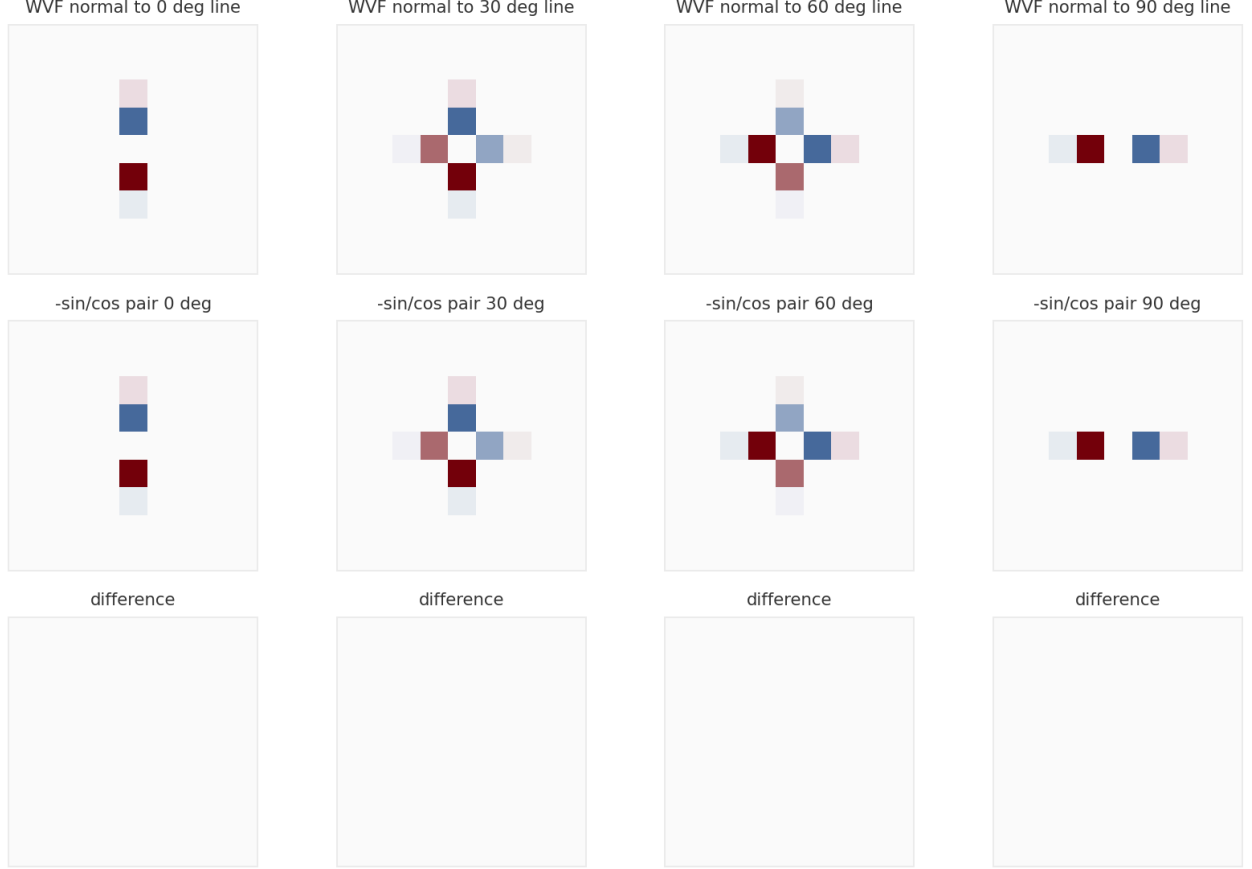


Figure 1: A numerical check of Equation 2.4 for the local Taylor/WVF derivative rows. The first row shows derivative kernels obtained by rotating the WVF coordinates to the normal of each line. The second row shows the same kernels reconstructed from one isotropic G_x and G_y pair. The third row shows the difference. The maximum row-weight error over the 36-angle bank is 6.66×10^{-15} .

3 The Standard LF Formulation

The standard LF response at pixel x and line angle θ can be written as a weighted sum of normal WVF responses at virtual positions along that line. With rounded virtual positions, define

$$q_{j(\theta)} = \text{round}(jv_\theta), \quad j \in \{-m, \dots, m\}. \quad (3.1)$$

For the standard LF line direction,

$$v_\theta = (\cos \theta, \sin \theta)^\top. \quad (3.2)$$

Then the standard LF response is

$$R_{\text{std}(x,\theta)} = \sum_{j=-m}^m w_j D_{n,\theta} I(x + q_{j(\theta)}). \quad (3.3)$$

This is the formulation that appears expensive. It appears to call for an orientation-indexed WVF normal derivative estimator $D_{n,\theta}$ at each virtual point $x + q_{j(\theta)}$. The next step is simply to substitute the isotropic identity from Equation 2.4 into Equation 3.3.

4 The Isotropic-WVF LF Rewrite

Substituting Equation 2.4 into Equation 3.3 gives

$$R_{\text{std}(x,\theta)} = \sum_{j=-m}^m w_j \left(-\sin \theta G_x(x+q_{j(\theta)}) + \cos \theta G_y(x+q_{j(\theta)}) \right). \quad (4.1)$$

The trigonometric factors do not depend on j , so they can be pulled outside the line sum.

$$R_{\text{std}(x,\theta)} = -\sin \theta \sum_{j=-m}^m w_j G_x(x+q_{j(\theta)}) + \cos \theta \sum_{j=-m}^m w_j G_y(x+q_{j(\theta)}). \quad (4.2)$$

Define the oriented line-smoothing operator

$$S_{\theta[f]}(x) = \sum_{j=-m}^m w_j f(x + q_{j(\theta)}). \quad (4.3)$$

Then Equation 4.2 becomes

$$R_{\text{std}(x,\theta)} = -\sin \theta S_{\theta[G_x]}(x) + \cos \theta S_{\theta[G_y]}(x). \quad (4.4)$$

This is the clean simplified form. The WVF is now only two isotropic derivative images. The orientation-dependent part is only line smoothing of those two derivative images, followed by the scalar $-\sin \theta, \cos \theta$ normal projection. This is the correction that matters. If θ names the line, the derivative direction is the other one, perpendicular to the line.

5 Kernel-Level Equivalence

The same statement can be written as a kernel identity. Let K_x and K_y be the two isotropic WVF derivative kernels. Let L_θ be the discrete line-smoothing kernel formed by the offsets $q_{j(\theta)}$ and weights w_j . Then the standard LF kernel for angle θ is

$$K_{\text{std}(\theta)} = L_\theta * (-\sin \theta K_x + \cos \theta K_y). \quad (5.1)$$

Here $*$ denotes the discrete composition produced by shifting the derivative kernel to each virtual line position and summing the shifted weights. This is exactly what the fused LF kernel builder does. It sums over line positions and WVF neighbor positions, but Equation 5.1 says those sums can be interpreted as line smoothing applied after one fixed isotropic derivative pair.

Line-normal LF kernels match the isotropic-WVF line-smoothed kernels

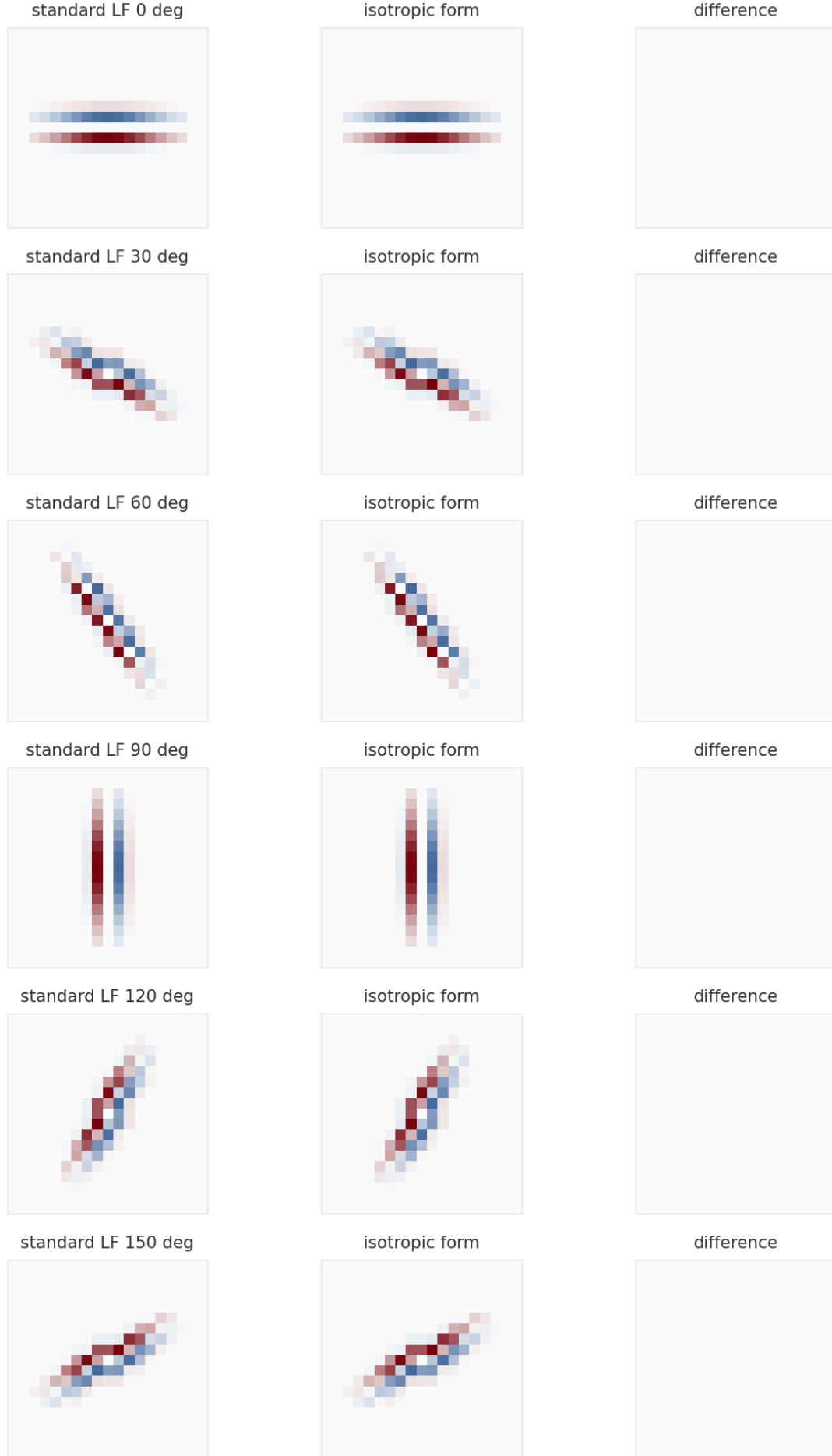


Figure 2: Standard line-normal LF kernels compared with kernels rebuilt from one isotropic $\frac{G_x}{G_y}$ pair plus oriented line smoothing. The maximum absolute kernel difference over the full 36-angle bank is 7.49×10^{-16} .

The practical interpretation is subtle. This derivation does not prove that the exact LF can be evaluated by only two image convolutions. It proves that the WVF portion can be evaluated by two isotropic convolutions. The line-smoothing operator S_θ still depends on θ . Therefore an exact implementation still needs orientation-dependent line smoothing, unless that line smoothing is accelerated by another method.

6 Comparison to the Standard LF

The standard and isotropic forms are summarized in Table 1. Both produce the same signed LF response when the WVF support is fixed and isotropic. The difference is where the orientation dependence is placed.

View	Expression	Interpretation
Standard LF	$\sum_j w_j D_{n,\theta} I(x + q_{j(\theta)})$	Evaluate the WVF normal derivative along the candidate line.
Isotropic-WVF LF	$-\sin \theta S_{\theta[G_x]}(x) + \cos \theta S_{\theta[G_y]}(x)$	Compute one fixed $\frac{G_x}{G_y}$ pair, then line-smooth and project onto the line normal.
Structure tensor	$n_\theta^\top J(x) n_\theta$	Square the derivative fields and maximize local second-moment energy. This is related but not the same LF response.

Table 1: Comparison between the standard LF expression, the exact isotropic-WVF rewrite, and the structure-tensor energy formulation.

The numerical comparison in Figure 3 uses BIPED RGB008 resized to a maximum side length of 128 pixels, with $m = 7$, $N_p = 15$, degree 4, circular WVF support, and 36 sampled line angles. The standard line-normal LF response stack and the isotropic-WVF response stack agree to floating-point precision. The maximum response difference is 2.00×10^{-15} , the relative response RMSE is 1.63×10^{-14} , and the gated argmax orientation agreement is exactly 1.0.

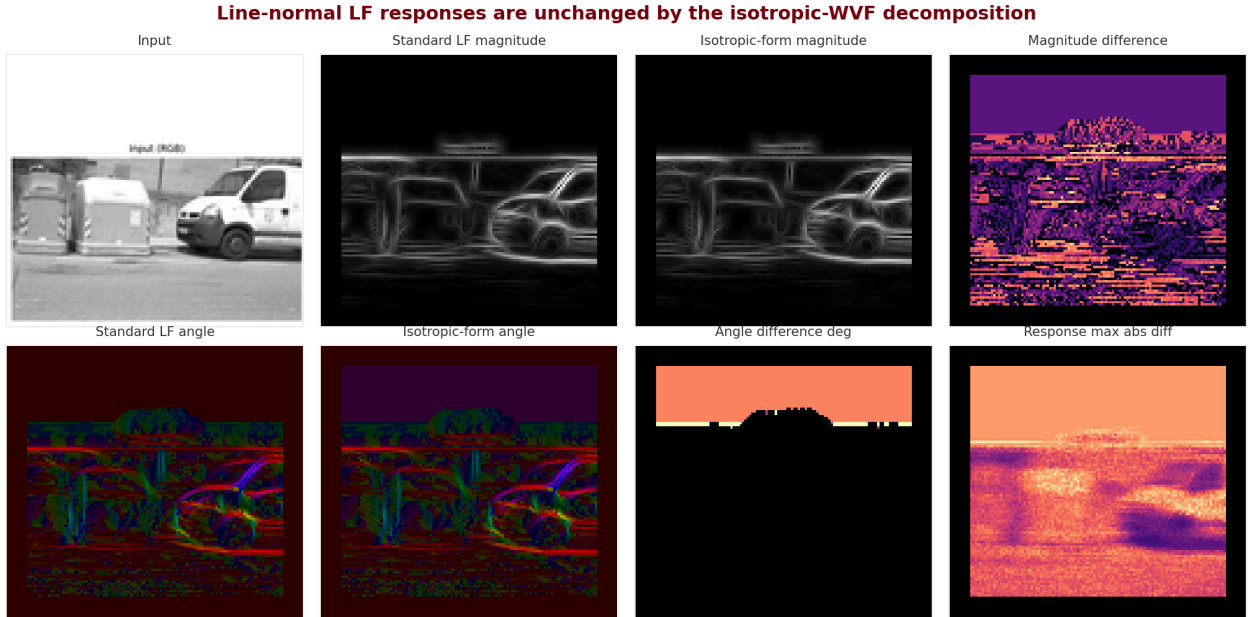


Figure 3: Image-level comparison between the standard LF response stack and the isotropic-WVF rewrite. The magnitude, angle, and response differences are at numerical roundoff.

7 What This Simplifies

The derivation moves the LF from an orientation-indexed WVF view to an isotropic derivative-plus-line-smoothing view. The exact response is unchanged, but the mathematical structure is cleaner.

The expensive object is no longer a WVF bank. It is an oriented line-smoothing bank applied to two derivative images. That is a better target for simplification. Line smoothing has more exploitable structure than repeated local polynomial fitting. It may be accelerated by separable line kernels, sparse stencil gathering, integral-image style sums for unweighted lines, or a low-rank angular basis. Those are implementation choices. The core mathematical point is that they should operate on G_x and G_y , not on a re-estimated anisotropic WVF.

This also explains why the structure tensor did not reproduce LF + spline. The tensor replaces the signed line-smoothed response in Equation 4.4 with squared local energy. It answers a different question. The isotropic-WVF LF rewrite keeps the signed linear evidence and therefore keeps the behavior of the standard LF.

8 The Normal-Indexed Variant

There is one geometric caveat worth making explicit. The formulas above use θ as the line direction. If instead a code path uses φ as the normal direction, then the same response can be written by rotating the smoothing direction by 90 degrees:

$$v_\varphi = (-\sin \varphi, \cos \varphi)^\top. \quad (8.1)$$

The equivalent normal-indexed form is

$$R_{\text{std}(x,\varphi)} = \cos \varphi S_{v_\varphi}[G_x](x) + \sin \varphi S_{v_\varphi}[G_y](x). \quad (8.2)$$

This is not a different filter. It is the same line-normal LF with the angle variable renamed from line direction to normal direction. What is not correct is using the same vector for both roles after deciding that θ names the line. If θ names the line, the WVF projection must use $-\sin \theta$ and $\cos \theta$.

9 Recommendation

The next simplification should be based on Equation 4.4. It is exact for the line-normal LF under an isotropic WVF and it gives a much better computational target than the structure tensor. The immediate implementation experiment should compute G_x and G_y once, apply the rounded line smoothing operator to each field for each line angle, and verify equality against a fused implementation of the same line-normal LF at full resolution. After that equality test is locked down, the remaining work is to approximate or accelerate only the oriented line-smoothing bank.

Bibliography

- [1] H. Bagan and Y. Wang, “Wide View Filter for Edge Detection in Underwater Images,” *IEEE Access*, vol. 9, pp. 152224–152238, 2021, doi: 10.1109/ACCESS.2021.3126787.
- [2] H. Bagan and Y. Wang, “Line Filter: Edge Detection on the Basis of a Local Polynomial Model and Virtual Expansion Points,” *IEEE Access*, vol. 11, pp. 49684–49700, 2023, doi: 10.1109/ACCESS.2023.3277440.